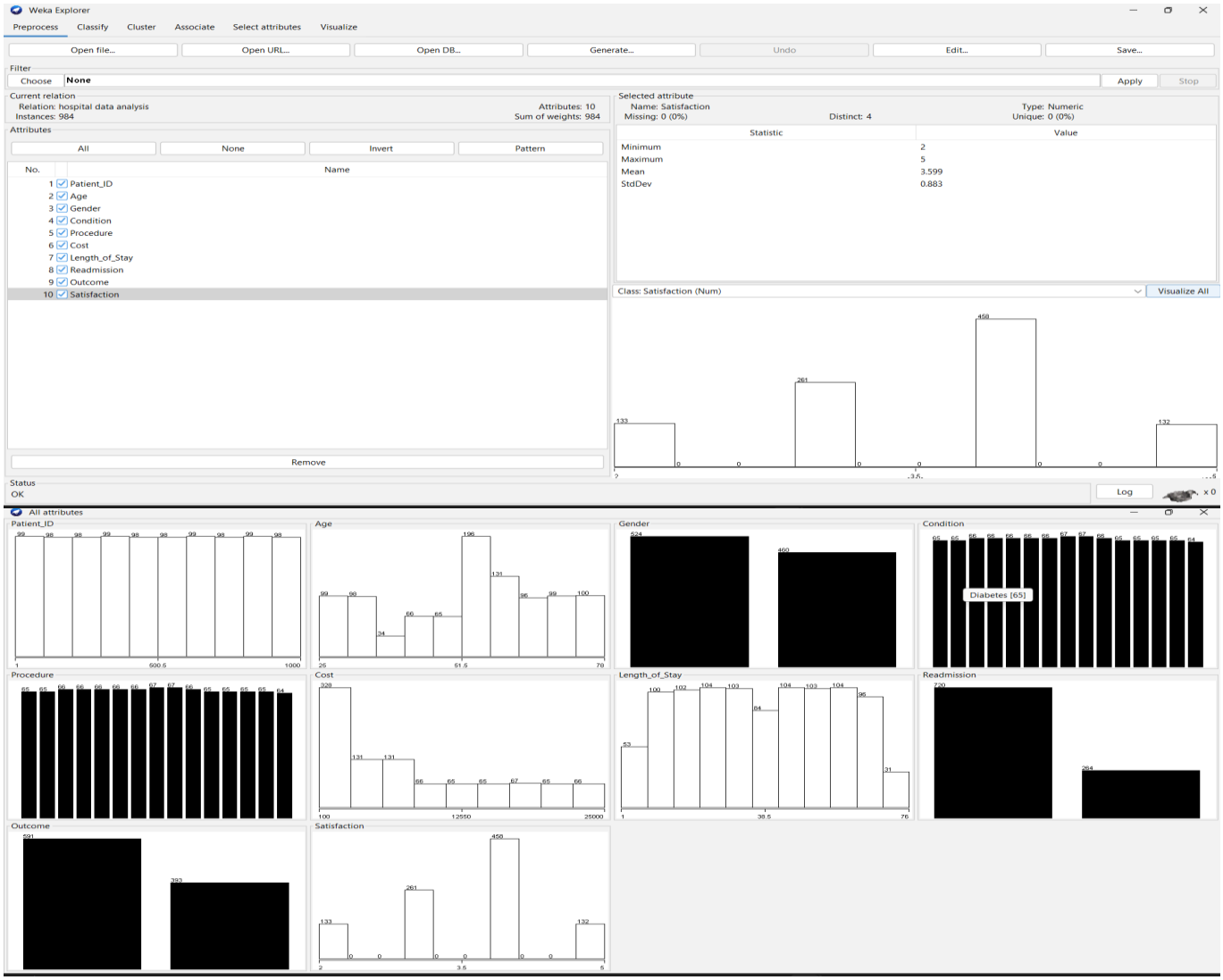
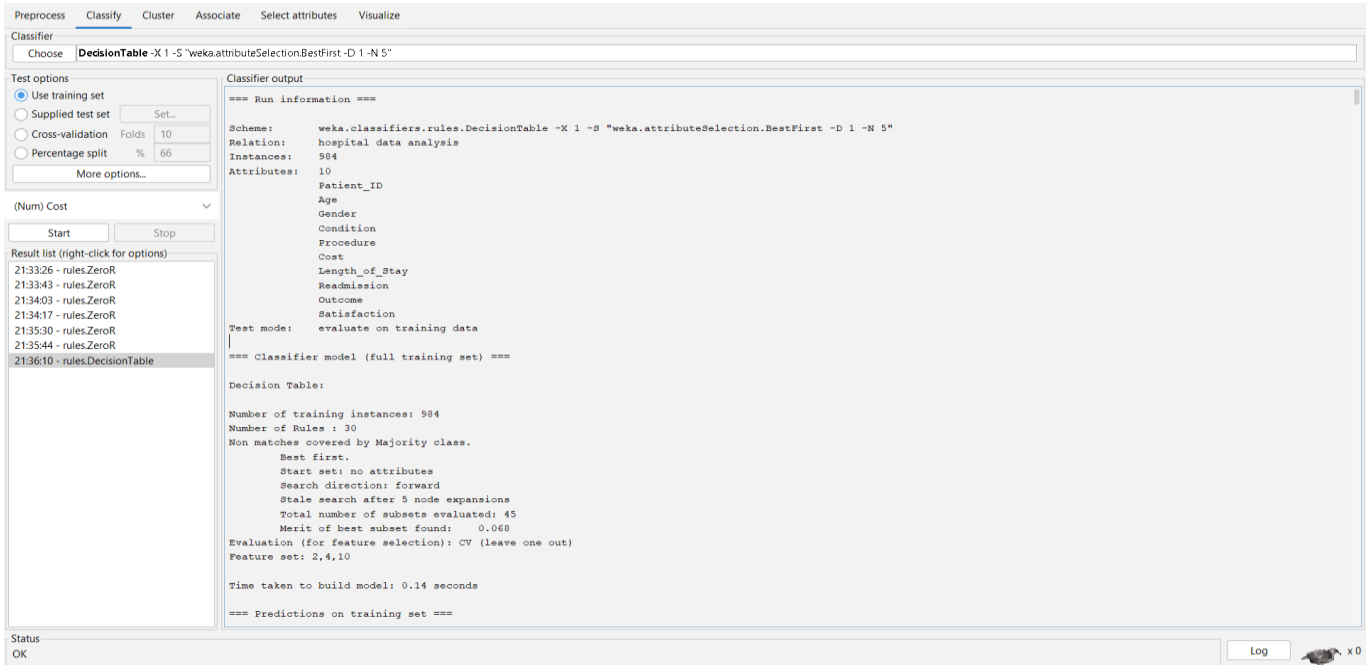
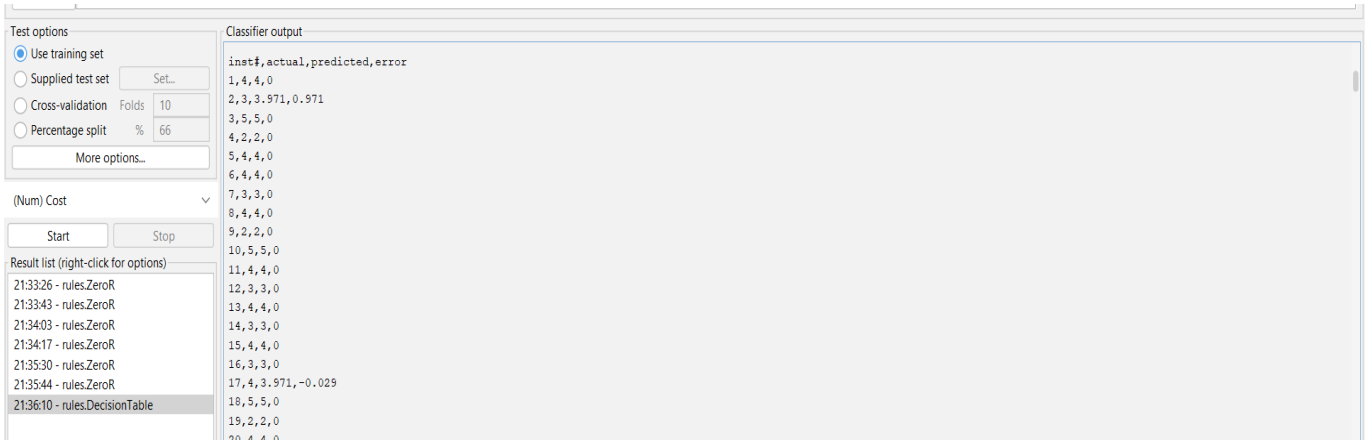
EXPERIMENT NO : 06

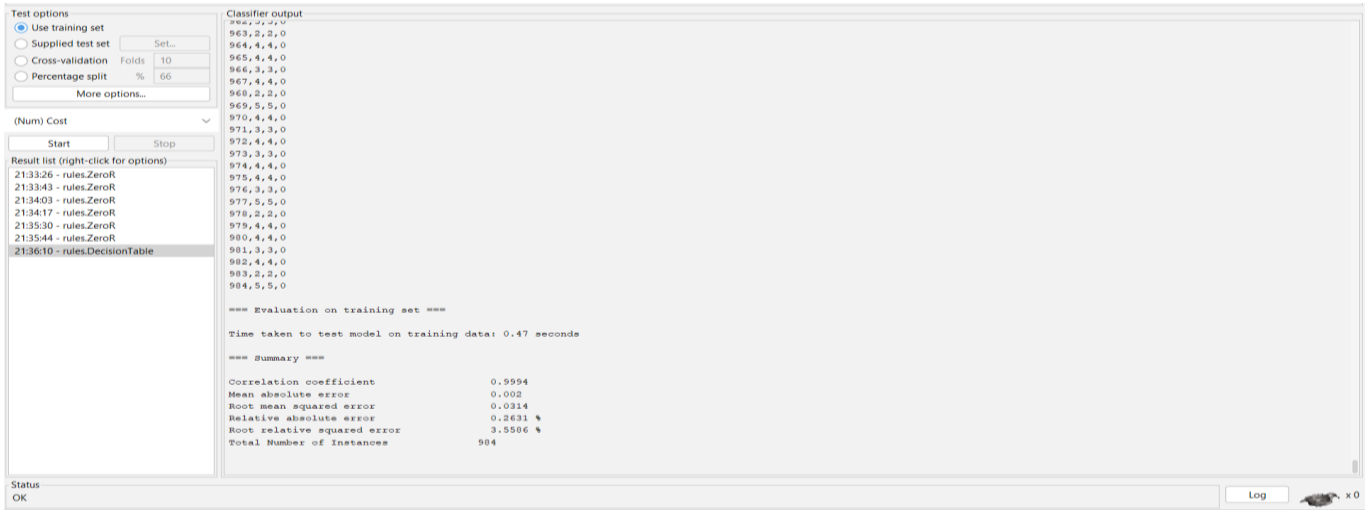
1. Preprocessing



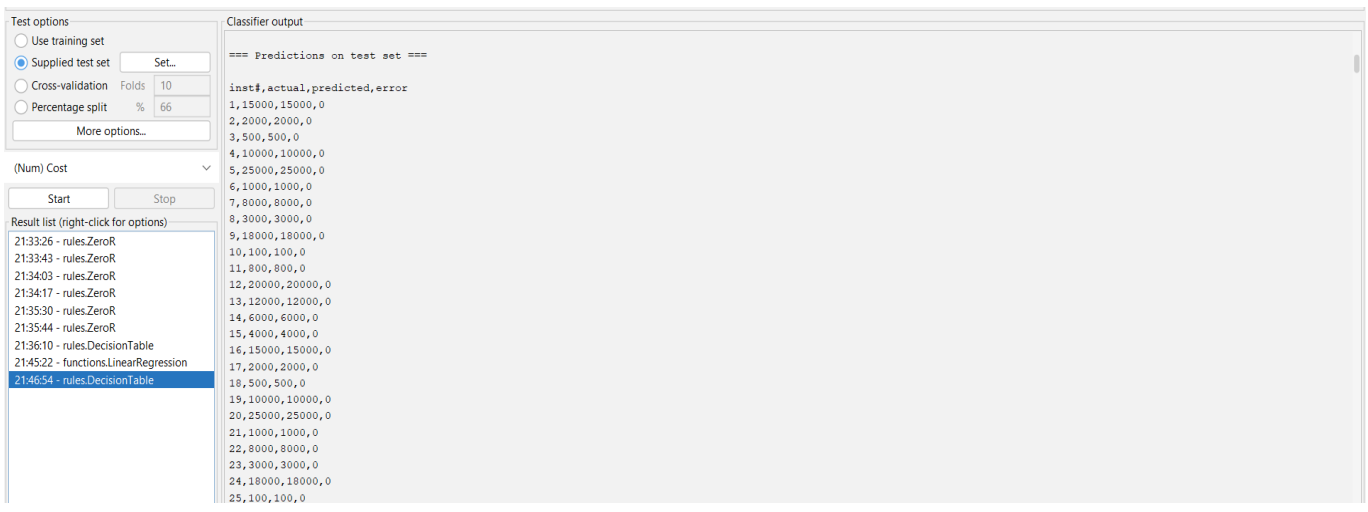
2.Classification(Decision Table Classifier)

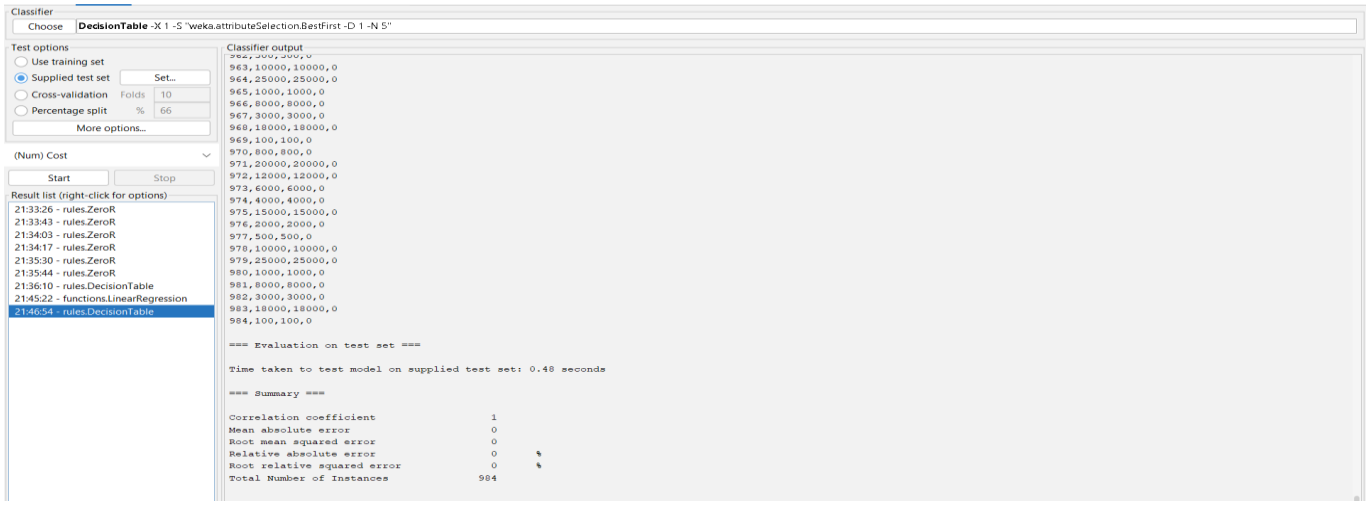



Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
Classifier output
inst#,actual,predicted,error
1,4,4,0
2,3,3.971,0.971
3,5,5,0
4,2,2,0
5,4,4,0
6,4,4,0
7,3,3,0
8,4,4,0
9,2,2,0
10,5,5,0
11,4,4,0
12,3,3,0
13,4,4,0
14,3,3,0
15,4,4,0
16,3,3,0
17,4,3.971,-0.029
18,5,5,0
19,2,2,0

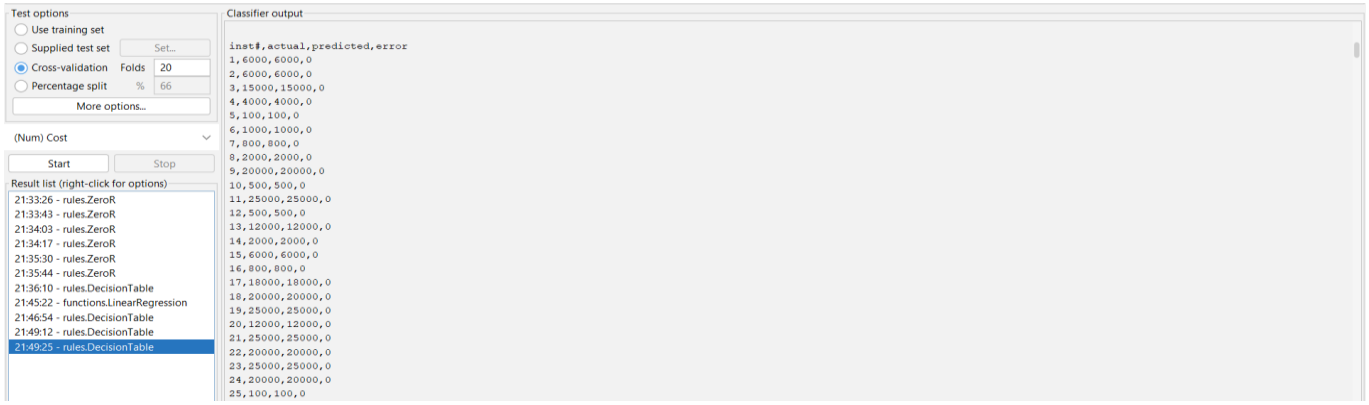

Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
Classifier output
963,2,2,0
964,4,4,0
965,4,4,0
966,3,3,0
967,4,4,0
968,2,2,0
969,5,5,0
970,4,4,0
971,3,3,0
972,4,4,0
973,3,3,0
974,4,4,0
975,4,4,0
976,3,3,0
977,5,5,0
978,2,2,0
979,4,4,0
980,4,4,0
981,3,3,0
982,4,4,0
983,2,2,0
984,5,5,0
=== Evaluation on training set ===
Time taken to test model on training data: 0.47 seconds
=== Summary ===
Correlation coefficient 0.9994
Mean absolute error 0.002
Root mean squared error 0.0314
Relative absolute error 0.2631 %
Root relative squared error 3.5586 %
Total Number of Instances 984
Status
OK
Log
x 0


Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
Classifier output
=== Run information ===
Scheme: weka.classifiers.rules.DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Relation: hospital data analysis
Instances: 984
Attributes: 10
Patient_ID
Age
Gender
Condition
Procedure
Cost
Length_of_Stay
Readmission
Outcome
Satisfaction
Test mode: user supplied test set: size unknown (reading incrementally)
=== Classifier model (full training set) ===
Decision Table:
Number of training instances: 984
Number of Rules : 15
Non matches covered by Majority class.
Best first.
Start set: no attributes
Search direction: forward
Stale search after 5 node expansions
Total number of subsets evaluated: 44
Merit of best subset found: 0
Evaluation (for feature selection): CV (leave one out)
Feature set: 4,6
Time taken to build model: 0.1 seconds
=== Predictions on test set ===


Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
Classifier output
=== Predictions on test set ===
inst#,actual,predicted,error
1,15000,15000,0
2,2000,2000,0
3,500,500,0
4,10000,10000,0
5,25000,25000,0
6,1000,1000,0
7,8000,8000,0
8,3000,3000,0
9,18000,18000,0
10,100,100,0
11,800,800,0
12,20000,20000,0
13,12000,12000,0
14,6000,6000,0
15,4000,4000,0
16,15000,15000,0
17,2000,2000,0
18,500,500,0
19,10000,10000,0
20,25000,25000,0
21,1000,1000,0
22,8000,8000,0
23,3000,3000,0
24,18000,18000,0
25,100,100,0


Classifier
Choose
DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
Classifier output
963,10000,10000,0
964,25000,25000,0
965,1000,1000,0
966,8000,8000,0
967,3000,3000,0
968,18000,18000,0
969,100,100,0
970,800,800,0
971,20000,20000,0
972,12000,12000,0
973,6000,6000,0
974,4000,4000,0
975,15000,15000,0
976,2000,2000,0
977,500,500,0
978,10000,10000,0
979,25000,25000,0
980,1000,1000,0
981,8000,8000,0
982,3000,3000,0
983,18000,18000,0
984,100,100,0
=== Evaluation on test set ===
Time taken to test model on supplied test set: 0.48 seconds
=== Summary ===
Correlation coefficient 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
Total Number of Instances 984

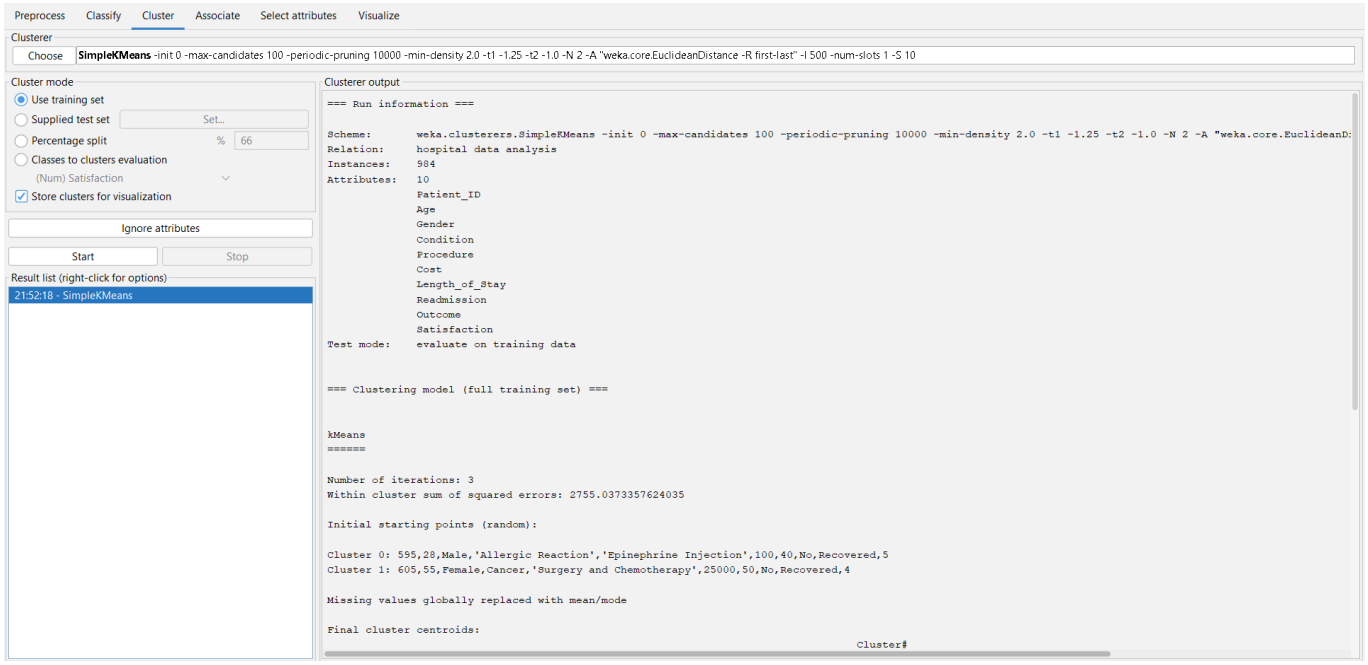

Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
20
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
21:49:12 - rules.DecisionTable
21:49:25 - rules.DecisionTable
Classifier output
=== Run information ===
Scheme: weka.classifiers.rules.DecisionTable -X 1 -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Relation: hospital data analysis
Instances: 984
Attributes: 10
Patient_ID
Age
Gender
Condition
Procedure
Cost
Length_of_Stay
Readmission
Outcome
Satisfaction
Test mode: 20-fold cross-validation
=== Classifier model (full training set) ===
Decision Table:
Number of training instances: 984
Number of Rules : 15
Non matches covered by Majority class.
Best first.
Start set: no attributes
Search direction: forward
Stale search after 5 node expansions
Total number of subsets evaluated: 44
Merit of best subset found: 0
Evaluation (for feature selection): CV (leave one out)
Feature set: 4,6
Time taken to build model: 0.02 seconds
=== Predictions on test data ===

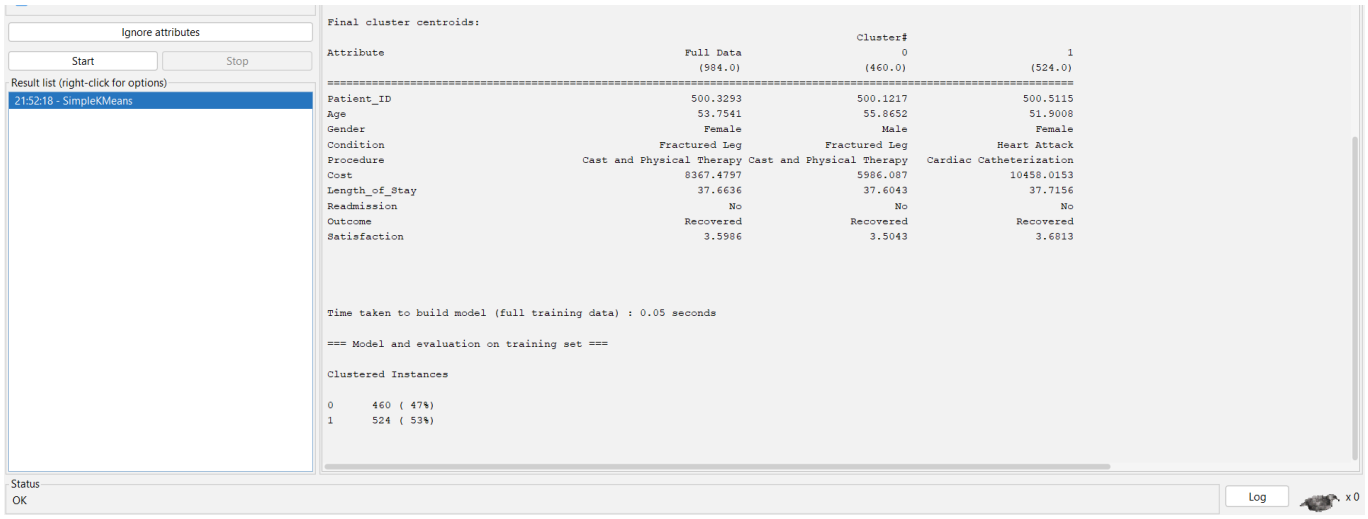

Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
20
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
21:49:12 - rules.DecisionTable
21:49:25 - rules.DecisionTable
Classifier output
inst#,actual,predicted,error
1,6000,6000,0
2,6000,6000,0
3,15000,15000,0
4,4000,4000,0
5,100,100,0
6,1000,1000,0
7,800,800,0
8,2000,2000,0
9,20000,20000,0
10,500,500,0
11,25000,25000,0
12,500,500,0
13,12000,12000,0
14,2000,2000,0
15,6000,6000,0
16,800,800,0
17,18000,18000,0
18,20000,20000,0
19,25000,25000,0
20,12000,12000,0
21,25000,25000,0
22,20000,20000,0
23,25000,25000,0
24,20000,20000,0
25,100,100,0


Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
20
Percentage split
%
66
More options...
(Num) Cost
Start
Stop
Result list (right-click for options)
21:33:26 - rules.ZeroR
21:33:43 - rules.ZeroR
21:34:03 - rules.ZeroR
21:34:17 - rules.ZeroR
21:35:30 - rules.ZeroR
21:35:44 - rules.ZeroR
21:36:10 - rules.DecisionTable
21:45:22 - functions.LinearRegression
21:46:54 - rules.DecisionTable
21:49:12 - rules.DecisionTable
21:49:25 - rules.DecisionTable
Classifier output
25,8000,8000,0
26,4000,4000,0
27,3000,3000,0
28,500,500,0
29,18000,18000,0
30,20000,20000,0
31,4000,4000,0
32,500,500,0
33,4000,4000,0
34,800,800,0
35,12000,12000,0
36,6000,6000,0
37,8000,8000,0
38,800,800,0
39,100,100,0
40,100,100,0
41,12000,12000,0
42,1000,1000,0
43,800,800,0
44,100,100,0
45,25000,25000,0
46,100,100,0
47,20000,20000,0
48,1000,1000,0
49,2000,2000,0
=== Cross-validation ===
=== Summary ===
Correlation coefficient 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
Total Number of Instances 984
Status
OK
Log
x 0

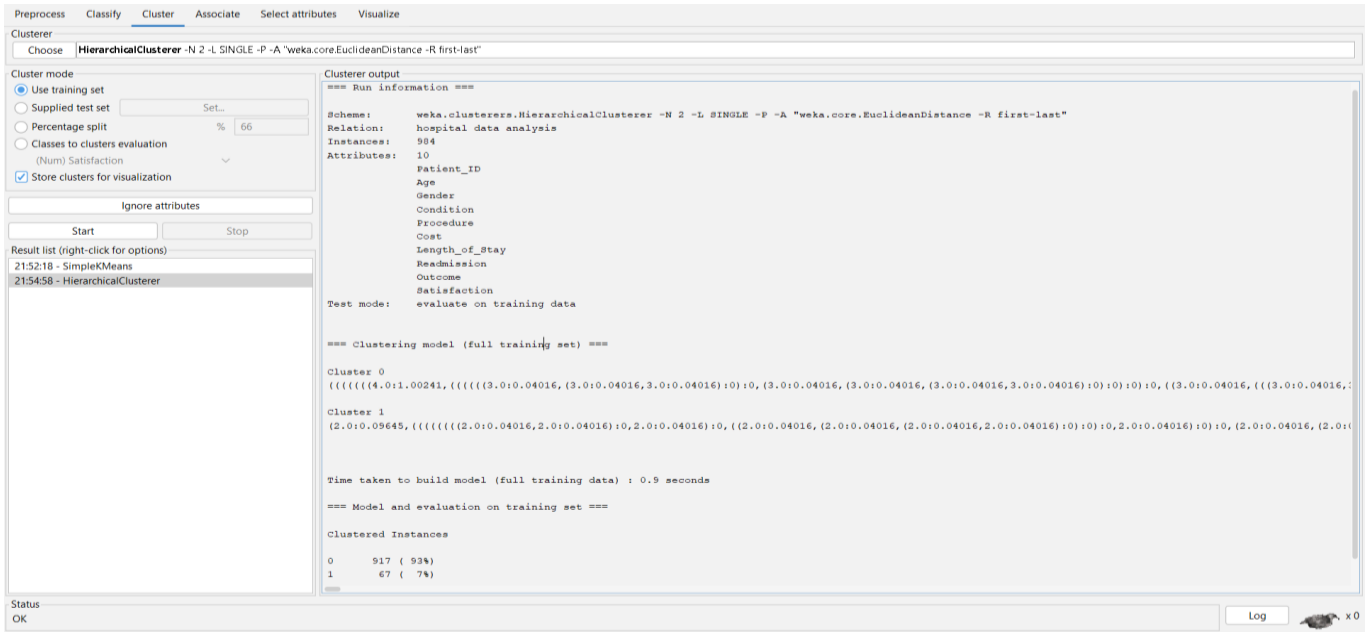
3.Clustering

- K Means Cluster





- Hierarchical Cluster



4. Association

